# 1 User Manual

## 1.1 Introduction

This manual describes the operation and functionality of the field-monitor embedded projects developed using the ATmega328P microcontroller. The system integrates three sensors LM35 temperature sensor, PIR motion detector, and soil moisture sensor and displays real-time readings on a 16×2 LCD. Users can switch between modes using push buttons, and the system provides visual and audible alerts when abnormal conditions are detected.

## 1.2 System Overview

The embedded system continuously monitors environmental parameters according to mode and provides feedback through an LCD, LEDs, and a buzzer. The system architecture consists of:

- ATmega328P microcontroller running at 16 MHz
- LM35 analog temperature sensor
- PIR motion sensor
- Soil moisture sensor (analog)
- Three push buttons for mode selection
- 16×2 LCD display
- Alert system: Red LED (PC1), Blue LED (PC2), Passive buzzer (PC4)

The system operates in three modes:

1. **LM35 Mode** – Displays temperature and triggers alerts for high/low thresholds.
2. **PIR Mode** – Detects motion and activates alarm when movement is present.
3. **Moisture Mode** – Displays soil moisture percentage and raw ADC value.

## 1.3 Hardware Requirements

The following hardware components are required for proper operation:

- ATmega328P microcontroller
- LM35 temperature sensor
- PIR motion sensor module

- Soil moisture sensor module
- 16×2 LCD
- Push buttons connected to PD3, PD4, PD5 with internal pull-ups
- Red LED (PC1), Blue LED (PC2)
- Passive buzzer (PC4)
- 5 V power supply
- Breadboard, wires, resistors

## 1.4 Software Requirements

To compile and upload the firmware, the following software is required:

- AVR-GCC toolchain
- FY-AVRISP MKII for flashing the microcontroller
- Development environment such as Microchip Studio or Arduino IDE
- Project driver libraries:
  - `lcd.h`, `lcd.c`
  - `adc.h`, `adc.c`
  - `pir.h`, `pir.c`
  - `moisture.h`, `moisture.c`

## 1.5 Installation

### *1.5.1 Hardware Setup*

1. Connect the LCD according to the pin mapping defined in `lcd.h`.
2. Connect the LM35 sensor to ADC0 (PC0).
3. Connect the PIR sensor to PD2.
4. Connect the soil moisture sensor to A5.
5. Wire push buttons to PD3, PD4, and PD5 with pull-up resistors enabled.
6. Connect LEDs to PC1 and PC2, and the buzzer to PC4.
7. Power the system using a stable 5 V supply.

### *1.5.2 Software Installation*

1. Open the project in your AVR development environment.
2. Ensure all driver files are included in the project.
3. Compile the code using compiler.
4. Upload the generated HEX file to the ATmega328P using AVR compatible programmer.

## 1.6 Usage

### *1.6.1 System Startup*

Upon powering the system, the LCD displays:

```
Sensor Board
Starting...
```

The system then enters LM35 Mode by default.

### *1.6.2 Mode Selection*

The three push buttons allow switching between modes:

- PD3 – LM35 Temperature Mode
- PD4 – PIR Motion Mode
- PD5 – Moisture Mode

A banner such as `>> LM35 Mode <<` is shown when switching modes.

### *1.6.3 Mode Descriptions*

**LM35 Temperature Mode**

- Displays temperature with one decimal precision.
- If temperature $\geq 35°C$ or $\leq 10°C$:
  - Red LED turns on
  - Buzzer plays a siren tone

**PIR Motion Mode**

- Displays "Motion: YES!!" when movement is detected.
- Motion triggers red LED and buzzer alarm.

**Moisture Mode**

- Displays soil moisture percentage and raw ADC value.
- If moisture $< 30\%$:
  - Red LED and buzzer activate
- Otherwise, blue LED indicates normal condition.

### 1.7 Troubleshooting

**LCD Not Displaying**

- Check power and contrast pin (VO).
- Ensure `lcd_init()` is executed.

**Buttons Not Responding**

- Verify wiring to PD3–PD5.
- Ensure internal pull-ups are active.

**Incorrect Sensor Readings**

- LM35: verify stable 5 V supply.
- PIR: allow 30–60 seconds warm-up.
- Moisture sensor: check for corrosion or loose wires.

**Buzzer Always On**

- Check if alert thresholds are constantly triggered.
- Verify sensor connections.

**Mode Switching Issues**

- Ensure buttons are not shorted.
- Confirm debounce logic is functioning.